

Deepak Mallampati

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Kent, OH (EST) -- US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) with strong API integration depth, POC-building experience across multiple AI providers, and proven ability to translate complex AI concepts for diverse technical and non-technical audiences. Experienced building production AI systems, designing custom proof-of-concepts, and enabling customers and teams to adopt AI tools. Python and TypeScript proficient; EST timezone (Ohio). Excited about voice AI as the next frontier of AI interfaces and eager to bring production AI engineering skills to Deepgram's enterprise technical pre-sales function.

CORE COMPETENCIES

Technical POC Development • API Integration (REST, multi-provider) • Python & TypeScript • LLM & AI System Design • RAG Pipeline Design • Demo & Prototype Engineering • Customer Technical Enablement • Docker & Cloud-Ready Deployment • Technical Documentation • Multimodal AI Integration

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting | USA

- Built **production AI proof-of-concepts and systems** for healthcare clients: RAG systems, agentic LLM pipelines, and predictive ML models -- demonstrating the POC-to-production capability central to pre-sales success.
- Designed **RAG systems** for clinical knowledge retrieval; improved contextual retrieval precision ~**35%**, achieved **>90%** retrieval-grounded response alignment.
- Developed **agentic LLM workflows** for multi-step healthcare automation; improved agent stability ~**25%** and reduced hallucinations **>30%**.
- Packaged inference services as **FastAPI/Flask RESTful APIs** with Docker; reduced post-deployment defects ~**30%** -- demonstrating API-first product delivery.
- Created technical documentation, system-limitation guides, and evaluation audit trails for non-technical stakeholders.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Applied NLP, Video Analytics | Hyderabad, India

- Built transformer-based video summarization system; reduced manual review time **60-70%**, achieved ~**85%** alignment with human-curated highlights.
- Deployed document Q&A with RAG for non-technical staff; raised contextual accuracy **>85%**.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Enterprise ERP | Hyderabad, India

- Optimized SQL/T-SQL stored procedures; reduced query latency ~**25%**.
- Built CI/CD pipelines with Jenkins; reduced deployment errors **>30%**.

SELECTED PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Translation shipped, Chrome Web Store

Independently built and shipped production browser extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-section capture pipeline, multi-provider payload normalization (WebP vs. JPEG handling), 7-day cache, 29-site support. Demonstrates multi-provider API integration and independent product delivery.

TypeScript, Manifest V3, Claude Sonnet API, GPT-4o mini, multimodal AI

Dream Decoder -- Multi-Provider AI Orchestration Platform

Full-stack FastAPI + React/TypeScript/Vite app integrating multiple AI APIs (Perplexity Sonar, Replicate image models); intermediate structured prompt transformation layers improved contextual alignment **~30%** and first-pass success rate **~25-30%**. Strong multi-provider API integration POC pattern.

FastAPI, React, TypeScript, Perplexity Sonar API, Replicate

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline with RAG-grounded validation, audit-ready reasoning traces, and schema-validated contracts. Complex AI POC built to production standards.

Python, LangChain, LlamaIndex, FastAPI, Docker, RAG

EDUCATION

Master's Degree -- Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: Python, TypeScript, FastAPI, Flask, React, SQL

LLM & AI: LangChain, LlamaIndex, RAG, agentic workflows, prompt engineering, evaluation pipelines, OpenAI API, Claude API

ML: scikit-learn, XGBoost, PyTorch, HuggingFace Transformers, embeddings, vector search

Infra: Docker, Jenkins, Grafana, CI/CD, RESTful APIs, observability/logging, cloud-ready deployment

Domains: API-First AI Products, Healthcare AI, Multimodal AI, Technical Pre-Sales Enablement